

Delhi Public School

Subject: Physics | Grade: 10th | Class: ____

Time: 2 Hours

Maximum Marks: 11

Name: _____

Roll No: _____

Section: _____

General Instructions:

1. Read all questions carefully before answering.
2. Write neatly and legibly.

Section A

All questions are compulsory. Choose the most appropriate option for each question. Each question carries 1 mark.

Q1. A concave mirror produces a magnification of -1 when the object is placed at which of the following positions? [Easy] [1]

- a. At infinity
- b. At the focus
- c. At the center of curvature
- d. Between the focus and the pole

Q2. The defect of vision in which a person can see nearby objects clearly but cannot see distant objects distinctly is known as: [Easy] [1]

- a. Myopia
- b. Hypermetropia
- c. Presbyopia
- d. Astigmatism

Q3. Two resistors of resistances 4 ohms and 6 ohms are connected in parallel to a voltage source. What is the equivalent resistance of this combination? [Moderate] [1]

- a. 10 ohms
- b. 2.4 ohms
- c. 1.2 ohms
- d. 5 ohms

Q4. The strength of the magnetic field inside a long current-carrying straight solenoid is: [Moderate] [1]

- a. More at the ends than in the center
- b. Minimum in the middle
- c. Same at all points
- d. Varying from one point to another

Q5. A convex lens of focal length 20 cm forms a real and inverted image of the same size as the object. The object must have been placed at a distance of: [Hard] [1]

- a. 10 cm
- b. 20 cm
- c. 40 cm
- d. 60 cm

Section B

Answer the following short answer questions in brief. Each question carries 2 marks.

Q6. State Joule's law of heating and write its mathematical expression. [Moderate] [2]

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Q7. Why do two magnetic field lines never intersect each other? [Moderate] [2]

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Q8. An object is placed perpendicular to the principal axis of a convex lens of focal length 10 cm. If the distance of the object from the lens is 15 cm, find the position and nature of the image formed.

[Hard]

[2]

— *End of Question Paper* —